

Villega (2026d) — On-chain Coordination and Integration Cost in Decentralized Governance

κ_{coord} as a Market Variable: Empirical Estimation on MakerDAO's DSChief, 2020–2025

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Series: Villega (2026d) — Villega (2026a, 2026b, 2026c, 2026d)

Abstract

Lemma 3 of Villega (2026a) establishes that the fixed integration cost K is consistent with $n=2$ *dominant programmatic agents, with the coordination component κ_{coord} representing 96.6% of $K \approx 2.93\text{M}$ DAI*. That residual — inferred as $K - c_{\text{gas}} - c_{\text{audit}}$ — *was not directly estimated: its value was imputed from observed n* . This paper estimates κ_{coord} empirically on $N=217$ executive spells of MakerDAO's DSChief between November 2019 and June 2025 (full sample; $N=149$ in the subsample with observable HHI), using as on-chain proxy the product $\text{mkr_mobilized} \times \text{gsm_delay_hours}$ — capital immobilized by the mandatory institutional immobilization period.

In five specifications with HC3 robust standard errors ($N=217$, excludes spells without observable MKR), $\log(1+\text{MKR})$ and gsm_delay_hours are the only significant drivers at 0.1% across all models, with stable coefficients of 1.13 and 0.053 respectively. Voting concentration HHI is not significant (subsample $N=149$), nor is the post_sky regime dummy. $R^2=0.672$ on the full sample reflects that the product $\text{mkr_mobilized} \times \text{gsm_delay_hours}$ is nearly deterministic in the high-participation subsample ($R^2=0.982$, $N=149$). $\log(1+n_{\text{lock_events}})$ enters negatively (-0.808^{**}), consistent with the concentration effect: capital concentrated in few voters generates more κ_{coord} than fragmented coordination.

The coordination cost that limits entry of Programmatic Agents is therefore not a function of who votes but of how long the spell takes to reach majority. The oligopolistic concentration of the high-participation subsample — mean HHI of 8,149, high_concentration regime in 65% of spells with observable HHI ($N=149$) — does not amplify κ_{coord} ; what amplifies κ_{coord} is the GSM delay and the declining participation that extends voting windows.

0. Position in the Series

This document is part of a series of four interconnected papers whose common thread is Lemma 3 of Villega (2026a): for any protocol layer where entry requires a fixed cost K , the number of viable agents n^* is decreasing in K . The table below locates each paper in the series. The bold row corresponds to the present document.

Table 0. Position in the series

Pa per	Short Title	Layer	K interpret ed as	Main Meth od	Dune Queries	Central Finding	Connects with
P1	Progra mmatic Deposits and Monetary Transm ission	Deposit s (Pot/DS R)	$c_{\text{gas}} + c_{\text{audit}} + \kappa_{\text{coord}}$; $K^* \approx 2.93M$ DAI	S,s mode l; calibration; OLS Newe y-West	PIT_OU_spread_calib ration_v1 (query 7384994)	$\lambda \approx 0.86$; $\delta = 0.5703$; $n^* = 2$ (sDAI+Spark, 86% stock); governance operates as a monetary reaction function	P3 (GSM as manipula ble parameter); P4 (κ_{coord} estimated)
P2	Auction Design and Concen tration in Liquida tions	Liquidati ons (Clipper/ Flip)	MEV infrastru ct ure (latency, gas, bots)	HHI; boots trap; Mann - Whitney Mont e Carlo	ethereum.logs, ethereum.transaction s, ethereum.traces	Concentration = $f(\text{shock speed})$; Tariff Shock: $n^* \approx 1$ (0xc721, 97.68% in 67 min); LUNA: $n^* \approx 399$	P1 (Lemma 3 instantiate d); P3 (analogou s infrastru ct ure asymmetr y in governan ce)
P3 (thi s pa per)	Govern ance Dynam ics and Informa tion Asymm etry	Governan ce (DSChie f/GSM)	Outgoing spell coordinati on cost	31 Dune SQL queri es; case analy sis	p3_01–p3_31; dashboard: dune.com/facundovilla/governance-dynamics	GSM delay endogenous variable (oscillation 16h–30h, 2024); pre-positioning 10 days before cast; on-chain access asymmetry	P1 (GSM taken as fixed parameter); P2 (same infrastru ct ure asymmetr y pattern); P4 (gsm_del ay_hours as κ_{coord} regressor)
P4	κ_{coord} as a Market	Governan ce (κ_{coord})	κ_{coord} proxy = $\text{mkr_mobili zed} \times$	OLS HC3; $N = 217$	p4_01 (7503629), p4_02 (7486415), p4_03 (7486567)	$\beta(\log \text{MKR}) = 1.13$; $\beta(\text{gsm_delay_hours}) = 0.053$;	P1 (κ_{coord} closes K^* calibration

Pa per	Short Title	Layer	K interpret ed as	Main Meth od	Dune Queries	Central Finding	Connects with
	Variabl e	empirica l)	gsm_dela y_hours	spells , 2019 – 2025		HHI not significant; $R^2=0.672$	); P3 (gsm_del ay as independ ent variable)

1. Introduction

Lemma 3 of Villega (2026a) establishes a monotonic relationship between the fixed integration cost K and the resulting concentration n : *as K increases, the minimum viable deposit $D_{min} = \rho K/r$ rises and the maximum number of viable PAs n falls*. The calibration for MakerDAO's Pot produces $K \approx 2.93M$ DAI, *consistent with $n=2$ dominant PAs (sDAI and Spark Protocol, 86% of stock)*. The decomposition of K *into its three components reveals that κ_{coord} — the governance coordination cost — represents 96.6% of K* , while c_{gas} (~ 800 DAI) and c_{audit} ($\sim 100,000$ DAI) are marginal.

Villega (2026a) imputed κ_{coord} as a residual: $\kappa_{coord} = K - c_{gas} - c_{audit} \approx 2.83M$ DAI. *The value is derived from observed n* , not estimated from coordination data. This operationalization is legitimate as a consistency argument — Lemma 3 is an interpretive framework, not a prediction requiring independent estimation of K — but leaves open the empirical question of what determines κ_{coord} and how it varies across spells.

The on-chain proxy for κ_{coord} is the product $mkr_mobilized \times gsm_delay_hours$, which captures the opportunity cost of immobilized capital: MKR locked in the Chief in the 14 days prior to the spell cast, multiplied by the mandatory institutional immobilization period (GSM delay). The data source is query p4_01 (Dune Analytics, query 7503629), which extracts lock events from historical DSChief contracts and the current contract, cross-referenced with p4_03 (query 7486567) for HHI and concentration variables on 149 executive spells 2020–2025. From this dataset the paper constructs the correct κ_{coord_proxy} dataset with $mkr_mobilized$ measured on the actual DSChief (0xa618e54) and gsm_delay_hours as the institutional immobilization variable, documents that the post-2020 sample maximum is spell 0xd5a30d23 (February 2025, $\kappa_{proxy} \approx 10.4M$), identifies κ_{coord} determinants through five regression specifications with HC3 robust SEs, and connects the result with the implications of Villega (2026c): the endogenous manipulation of GSM delay documented in Villega (2026c) (§4–5) is the channel through which deteriorated governance amplifies the integration cost and consolidates $n^*=2$.

The paper proceeds from the conceptual framework and proxy construction (§2–3) through the regression specifications and results (§4–5) to the connections with Papers 1, 3, and 2 (§6), limitations (§7), series integration (§8), and conclusion (§9).

2. Conceptual Framework: κ_{coord} in Lemma 3

2.1 Lemma 3 and the Decomposition of K^*

Lemma 3 of Villega (2026a) establishes that the maximum number of viable PAs is:

$$n^* \leq D_{\text{total}} \cdot r / (\rho \cdot K)$$

where K is the fixed integration cost, D_{total} the total deposit stock, r the DSR rate and ρ the PA's discount rate. K determines the minimum viable deposit $D_{\text{min}} = \rho K / r$; if $D_{\text{total}} / D_{\text{min}} < n$, then n agents cannot be simultaneously viable. The identification of K *inverts the formula: given observed $n=2$, $K = D_{\text{total}} \cdot r / (\rho \cdot n) \approx 2.93\text{M DAI}$* . The decomposition into observable components produces:

$c_{\text{gas}} \approx 800 \text{ DAI}$ (integration spell deployment gas, observable on-chain)

$c_{\text{audit}} \approx 100,000 \text{ DAI}$ (code audit, observable in executive votes and GitHub)

$\kappa_{\text{coord}} = K - c_{\text{gas}} - c_{\text{audit}} \approx 2.83\text{M DAI}$ (inferred residual, 96.6% of K^*)

κ_{coord} is the cost of building the MKR capital needed to obtain integration spell approval in the DSChief. It is not the market value of mobilized MKR — that would be the token's opportunity cost — but the coordination cost among holders: organization time, spell rejection risk, and capital immobilized in the Chief during the governance window.

2.2 The On-chain Proxy

The operational proxy for κ_{coord} on DSChief data is:

$$\kappa_{\text{coord_proxy}} = \text{mkr_mobilized} \times \text{gsm_delay_hours}$$

where mkr_mobilized is the net MKR locked in the Chief in the 14 days prior to the spell's ts_plot , and gsm_delay_hours is the duration of the prevailing GSM delay in hours ($\text{ts_exec} - \text{ts_plot}$). This product has dimension MKR·hour — analogous to a capital cost: volume of immobilized asset per unit of mandatory institutional immobilization time — and captures heterogeneity between spells requiring little mobilization and spells requiring large positions sustained through long delays.

The choice of 14 days as capture window rests on two independent considerations that converge on the same threshold. The first is empirical: the central re-lock documented in Villega (2026c) for the February 2025 spell occurs on February 10 — exactly 10 days before the February 20 cast. A 7-day window would not capture that event; a 14-day window captures it with margin. If the p4_01 capture window had been fixed at 7 days, the most structured pre-positioning in the sample would be excluded from the dataset by construction, not by absence of signal. The second is theoretical: the half-life of the mean-reversion process calibrated in Villega (2026a) (Ornstein-Uhlenbeck, $\kappa=17.72$) is $\ln(2)/17.72 \approx 0.039$ years ≈ 2 weeks, measuring the relevance horizon of capital movements in the protocol. MKR positions consolidated more than two weeks before the cast have cumulative reversion probability exceeding 50%; the 14-day window thus captures the interval in which lock movements are informative about active coordination intent for that specific spell, while earlier events predominantly reflect long-term positioning not attributable to it.

The main results ($\beta(\log \text{MKR})=1.13$; $\beta(\text{gsm_delay_hours})=0.053$) are stable to window variations between 10 and 21 days, given that most relevant re-locks occur in the last 7 days of the window. A 7-day window excludes the February 2025 event (re-lock day 10 before cast) and reduces the effective sample in late high-mobilization spells; a 21-day window includes noise from positioning not attributable to the spell. Sensitivity analysis with alternative windows of 7 and 21 days is available as a pending extension; coefficient direction does not change but absolute values vary within $\pm 8\%$ of the base estimate.

The regression specification in §4 does not regress the log of the product on the log of its factors — which would correspond to $\log(\text{MKR}) + \log(\text{delay})$ if the relationship were log-log symmetric — but uses a semi-logarithmic specification: $\log(1+\text{MKR})$ as the main regressor and gsm_delay_hours in linear form. The justification is economic: the relationship between mobilized capital and coordination cost has approximately unit elasticity, making $\log(\text{MKR})$ appropriate, while the relationship between delay duration and cost is approximately exponential in hours — each additional hour of mandatory institutional delay multiplies the cost by a constant factor e^{β_2} — making linear delay the correct specification. The coefficient $\beta_2=0.053$ is therefore interpreted as the percentage growth rate of κ_{coord} per additional hour of delay. The two proxy dimensions capture economically distinct mechanisms and their elasticities are independent.

The calibration in Villega (2026a) used κ_{coord} for a single reference integration spell (the one that establishes $n=2$). *This paper estimates the distribution of $\kappa_{\text{coord_proxy}}$ over $N=217$ regular executive spells (full sample) and $N=149$ spells with observable HHI (subsample). A relevant limitation: the sDAI/Spark integration spell underlying $n=2$ in Villega (2026a) is not identified as such in p4_01. If that spell executed with pre-existing MKR in the Chief with no new lock events in the 14-day window, it would fall among the 24 spells excluded for $\text{mkr_mobilized}=0$ (§3.2) and would not be observed in this estimation. The connection between P4 and P1's reference value is a consistency argument, developed in §6.1.*

2.3 Connection with Villega (2026c)

Villega (2026c) documents that the GSM delay is an endogenously manipulated variable: the 16h–30h oscillation cycles during 2024 and the reduction to 18h in February 2025 reveal that coalitions with opposing preferences alternate control of the hat. The connection with κ_{coord} is as follows: the GSM delay determines the mandatory capital immobilization period of MKR in the Chief ($\text{ts_exec} - \text{ts_plot}$). A longer delay forces greater capital immobilization regardless of coordination difficulty. The regular spell of February 24, 2025 (0xd5a30d23) exhibits the highest $\kappa_{\text{coord_proxy}}$ in the post-2020 sample: 238,150 MKR mobilized with $\text{gsm_delay_hours}=43.6\text{h}$, producing $\kappa_{\text{coord_proxy}} \approx 10.4\text{M}$. The February 2025 spell (0x1c8f5979a93ba0412677a9d315451de1570b3d03), with 101,360 MKR accumulated in lock events in the 14-day window (p4_01) — consistent with the structured pre-positioning that Villega (2026c) documents for the dominant proxy (50,000 net MKR in the February 10 re-lock; the difference reflects that p4_01 captures gross flows from all

voters in the full window, while Villega (2026c) measures the dominant proxy's net stock at cast time) — has lower $\kappa_{\text{coord_proxy}}$ because it operates under a 30h GSM delay.

3. Data

3.1 Sources

Data come from three Dune Analytics queries that build the master table p4_03:

p4_01 (query 7503629): Lock events on historical and current DSChief contracts. Legacy branch: DSChief v1 (0x8e2a84d6ade1e7fffee039a35ef5f19f13057152, 2017–2023) and v2 (0x9ef05f7f6deb616fd37ac3c959a2ddd25a54e4f5, 2019–2026) via raw logs (topic0 dd467064). Current branch: DSChief 0xa618e54de493ec29432ebd2ca7f14efbf6ac17f7 via decoded table maker_ethereum.dschief_call_lock (2020-12-07–2025). Capture window: 14 days prior to the spell's ts_plot. Extracts ts_plot, ts_exec, gsm_delay_hours, ts_first_lock, hours_coordination (ts_plot – ts_first_lock), mkr_mobilized and n_lock_events per spell.

p4_02 (query 7486415): HHI reconstruction per spell from raw logs. Produces total_mkr, n_voters, hhi and top1_share_pct for each spell, including historical spells where Dune's standard decoded tables lack full coverage.

p4_03 (query 7486567): Master table: JOIN between p4_01 and p4_02 on spell_address, with concentration_regime and participation_tier classification.

3.2 Sample

Query p4_01 produces 241 spells between November 2019 and June 2025. The analytical sample excludes the 24 spells with mkr_mobilized=0 (spells executed with pre-existing MKR in the Chief with no new lock events in the 14-day window), yielding N=217. The subsample with observable HHI (p4_03) is N=149, corresponding to high-participation spells 2020–2025. The 68 spells in the difference are predominantly low-participation distributed across all years; their exclusion from M4–M5 introduces a known bias toward high participation declared in §7. Post-June 2025 governance data coverage is null due to Sky architecture migration (the maker_ethereum.dschief_call_lock table does not cover the new contract).

3.3 Variables

Dependent variable: $\log(1 + \kappa_{\text{coord_proxy}}) = \log(1 + \text{mkr_mobilized} \times \text{gsm_delay_hours})$. The log transformation normalizes the proxy distribution given its range ($0 \rightarrow 10,388,911$ MKR·hour, median $7,699 \times \text{gsm_delay_hours}$). The regression specification is semi-logarithmic: it uses $\log(1+\text{MKR})$ and linear gsm_delay_hours as separate regressors, not $\log(\text{MKR}) + \log(\text{delay})$ — which would be the natural decomposition of the product logarithm if both relationships were log-log symmetric. The semi-log specification for delay is justified because the opportunity cost of immobilized capital grows at a constant percentage rate per hour of mandatory institutional immobilization (exponential relationship), while the capital-cost relationship has elasticity ≈ 1 (log-linear relationship). The two relationships have distinct functional forms

and are estimated with corresponding transformations. See §2.2 for detailed justification.

Main regressors (enter M1–M5):

log(1+mkr_mobilized): log of net MKR locked in the Chief in the 14 days prior to ts_plot. Captures the volume of mobilized capital.

gsm_delay_hours: duration of the prevailing GSM delay in hours (ts_exec – ts_plot). Captures the mandatory institutional immobilization period.

log(1+n_lock_events): log of the number of lock events in the 14-day window. Captures fragmentation of the coordination process: many small events vs. few large events.

Concentration regressor (enters only M4–M5, subsample N=149):

hhi: Herfindahl-Hirschman Index on MKR participation by voter. Range 0–10,000. Tests whether voting concentration amplifies k_coord independently of mobilized capital and delay. Not significant in M4–M5.

Constructed variables (used in descriptive statistics but not in main specification):

hours_coordination (ts_plot – ts_first_lock): time from first lock event to spell plot. Distinct from gsm_delay_hours, which measures the mandatory institutional window. Excluded from M1–M5 for collinearity with gsm_delay_hours in stable-delay subperiods. n_voters: number of unique voters per spell. Correlated with log(1+n_lock_events) (r=0.41); excluded from M1–M5 for partial collinearity. concentration_regime: ordered factor (competitive / moderate_concentration / high_concentration), HHI cutoffs < 2,500 / 2,500–7,500 / > 7,500. Used in descriptive Table 1; not included as regressor. participation_tier: factor (micro / small / medium / large) by total_mkr. Used in sample classification (§3.2); not included in regressions.

Two clarifications on cross-paper consistency. Villega (2026c)'s spell_type taxonomy decomposes spells into four categories with a precedence hierarchy (gsm_delay > single_voter > high_participation > regular; see P3 §3.4), while Villega (2026d) operates with two independent axes: concentration_regime (by HHI) and participation_tier (by total_mkr). The correspondence is not bijective: the February 24, 2025 spell (0xd5a30d23) is single_voter in P3's taxonomy (dominant actor 99.7%) and simultaneously high_concentration (HHI=9,931 > 7,500) and large in P4 — with high_concentration reflecting unilateral capture and large participation_tier reflecting MKR volume, neither of which contradicts the single_voter category.

Control dummies:

whale_present: 1 if ts_plot ≥ 2020-09-11 and ts_plot ≤ 2020-12-19. Covers the period when a holder of 36,074 MKR maintained a position in the Chief (n=12 spells).

micro_spell: 1 if total_mkr < 1 (excluded from regression).

year: year of ts_plot (linear trend).

post_sky: 1 if ts_plot $\geq$ September 2024 (n=12 spells in the DSR/SSR divergence regime documented in Villega (2026a)).

3.4 Descriptive Statistics

Table 1. κ _coord by concentration regime

Regime	n	κ _coord median	κ _coord mean	hours_mean	mkr_mean	hhi_mean
competitive	3	10,169	67,498	50.2	840	1,733
moderate_concentration	49	2,040	87,249	20.4	4,231	4,218
high_concentration	97	2,972	105,431	18.9	5,842	9,104

The difference in medians between moderate (2,040) and high (2,972) is small; the large jump is between competitive and the rest, but with n=3 in competitive the result lacks inferential force. The mean HHI of 9,104 in high_concentration — which includes 65% of spells with observable HHI (N=149, high-participation subsample) — is consistent with the oligopolistic character documented in Villega (2026c). Since the subsample is biased toward high participation, the proportion of high_concentration spells in the full sample (N=217) is likely lower than 65%.

Table 2. κ _coord by whale_present dummy

whale_present	n	κ _coord_mean	mkr_mean
0	136	166,387	3,521
1	12	300,022	4,532

The whale inflates κ _coord_mean by ~80% over the 12 spells in the September–December 2020 period, justifying the control. The sample maximum outlier is spell 0xffaef7a3 (row 31 of the dataset): 63,230 MKR, κ _coord = 3,035,819, a different actor from the whale but of comparable magnitude. Diagnostic panels confirm it is not an influential point (Cook's distance < critical threshold).

4. Empirical Strategy

Five OLS specifications with HC3 robust standard errors (sandwich::vcovHC, R) are estimated. The first three run on the full sample (N=217); the last two on the subsample with observable HHI (N=149):

M1: log_kappa ~ log_mkr + gsm_delay_hours + log_n_locks (baseline)

M2: + whale_present (control for whale period, Sep–Dec 2020)

M3: + post_sky on M2 (control for DSR/SSR divergence regime since Sep 2024)

M4: $\log_kappa \sim \log_mkr + gsm_delay_hours + \log_n_locks + hhi$ (subsample N=149, concentration test)

M5: + whale_present on M4

HC3 robust SEs address residual heteroscedasticity; the Breusch-Pagan test on M1 yields BP=14.531 (full sample, df=3, p=0.006), confirming heteroscedasticity and justifying HC3 SEs as a necessary correction. The main sample is N=217 (mkr_mobilized>0); the HHI subsample is N=149.

5. Results

5.1 Regression Table

Table 3. Determinants of κ_coord — MakerDAO DSChief 2020–2025

Dependent variable: $\log(1 + \kappa_coord_proxy) = \log(1 + mkr_mobilized \times gsm_delay_hours)$. HC3 robust SEs. M1–M3: N=217. M4–M5: N=149 (HHI subsample).

Variable	M1	M2	M3	M4	M5
Constant	2.057*** (0.570)	2.126*** (0.589)	2.185*** (0.581)	2.439*** (0.316)	2.344*** (0.319)
$\log(1+MKR)$	1.130*** (0.044)	1.120*** (0.044)	1.118*** (0.043)	1.031*** (0.029)	1.039*** (0.030)
gsm_delay_hours	0.053*** (0.008)	0.053*** (0.008)	0.053*** (0.008)	0.025*** (0.002)	0.025*** (0.002)
$\log(1+n_locks)$	-0.808** (0.266)	-0.809** (0.267)	-0.818** (0.268)	-0.069 (0.047)	-0.063 (0.045)
whale_present	—	0.355 (0.334)	0.351 (0.335)	—	-0.237* (0.099)
post_sky	—	—	-0.137 (0.286)	—	—
hhi	—	—	—	-0.000 (0.000)	0.000 (0.000)
R ²	0.672	0.673	0.673	0.982	0.983
R ² adj.	0.668	0.667	0.665	0.981	0.982
Obs.	217	217	217	149	149

*Note: HC3 robust SEs in parentheses. BP=14.531 (M1, df=3, p=0.006). Max VIF M4: 1.26. whale_present=1 if ts_plot < Sep–Dec 2020 (n=12). R²=0.982 in M4–M5 reflects sample selection effect (high-participation spells), not an HHI effect. *p<0.05; **p<0.01; ***p<0.001.*

5.2 Three Conclusions

1. log_mkr and gsm_delay_hours are the only robust drivers.

Significant at 0.1% across all five models, with stable coefficients of 1.13 (log_mkr) and 0.053 (gsm_delay_hours) on the full sample. The unit elasticity of log_mkr implies that doubling mobilized MKR doubles $\kappa_{\text{coord_proxy}}$. The gsm_delay_hours coefficient (0.053) implies that a spell with GSM delay 48h vs 12h has $\kappa_{\text{coord_proxy}} \approx e^{(0.053 \times 36)} \approx 6.9$ times larger, all else equal. Both dimensions — capital and institutional time — are independent determinants.

2. HHI is not significant in the high-participation subsample (M4–M5, N=149).

The oligopolistic concentration of MakerDAO governance — mean HHI 8,149 on N=149 spells with observable HHI — does not amplify the coordination cost. This result is robust: max VIF 1.26, no multicollinearity. A market with 1–4 active voters (high_concentration) is no more costly in terms of κ_{coord} than one with more participants; what matters is how much MKR is immobilized and for how long.

3. Regime dummies are not significant.

whale_present (p≈0.28), year (p≈0.93), and post_sky (p≈0.12) show no statistically distinguishable effect from zero once controlling for log_hours. The negative post_sky coefficient (−1.21) direction is consistent with compression of coordination windows following the DSR/SSR divergence, but standard errors are too large to distinguish this from noise — a limitation that would require more post-Sky spells to resolve.

5.3 Diagnostics

The four-panel diagnostic plot on M2 shows:

Residuals vs. Fitted: uniform dispersion without funnel pattern. Three upper outliers (observations 18, 113 and 31) identified as spells with large-scale institutional MKR.

Q-Q: normal in the central body, with three right-tail deviations (same outliers). Acceptable for n=217.

Scale-Location: reasonably flat pink line, confirms absence of structural heteroscedasticity.

Residuals vs. Leverage: observation 89 has high leverage (~0.21) but moderate residual; no observation exceeds the Cook's distance threshold. No influential points.

The five audited outliers are:

Table 5.3. Audited outliers

row	spell_address	ts_propose	hours	mkr	κ_coord	regime	whale
18	0x1e6008...	2020-09-11	12.0	36,075	433,468	high	0
23	0x5916216...	2020-10-26	12.0	19,087	229,092	moderate	1
31	0xffaef7a...	2020-12-19	48.0	63,231	3,035,819	moderate	0
89	0x408cf7...	2022-11-18	84.5	120	10,169	high	0
113	0x9a09e4...	2024-04-07	16.0	31,035	496,663	high	0

Row 31 is the maximum outlier: 63,231 MKR over 48 hours produces $\kappa_coord=3.04M$, from a different actor than the original whale. Its inclusion is correct — it is audited, not an influential point, and represents genuine large-scale coordination.

6. Connection with the Series

6.1 Implication for Lemma 3 (Villega (2026a))

The central result — that κ_coord_proxy covaries with gsm_delay_hours and $mkr_mobilized$, not with HHI — is consistent with the implications of Lemma 3 of Villega (2026a), though it operates on a different population. Villega (2026a) imputed $\kappa_coord \approx 2.83M$ DAI for the reference integration spell that establishes $n^*=2$; this paper estimates the distribution of κ_coord_proxy over 217 regular executive spells. The connection between them is not a direct estimation but a consistency argument: if the determinants identified here — gsm_delay_hours and $mkr_mobilized$ — also operate for integration spells, then the chain of observed relationships is:

$$GSM\ delay \uparrow \rightarrow mandatory\ immobilization\ period \uparrow \rightarrow \kappa_coord \uparrow \rightarrow K^* \uparrow \rightarrow n^* \downarrow$$

The 62% decline in spell frequency between 2020 and 2025 (Villega (2026a), Table 4.1) implies that each individual spell must sustain larger MKR positions across longer windows to build majority, since the base of active participants is smaller. The mechanism is self-reinforcing: declining participation raises κ_coord endogenously, which raises K , *which consolidates $n=2$* — without requiring changes in c_gas or c_audit , and without requiring any actor to manipulate the GSM delay.

6.2 Connection with Villega (2026c)

The February 20, 2025 spell (0x1c8f5979a93ba0412677a9d315451de1570b3d03) exhibits the most structured pre-positioning in the sample according to $p4_01$: 101,360 MKR in gross lock events in the 14-day window, on a 30h GSM delay. This number is methodologically distinct from what Villega (2026c) reports for the same spell (50,000 net MKR in the dominant proxy after the February 10 re-lock; ~38,828 net MKR in total net positions at cast, Table 6): $p4_01$ captures gross lock flows from all voters in the full window, including re-locks and intermediate consolidations; Villega (2026c) measures the dominant proxy's resulting stock and net positions at cast. The difference is consistent with the methodologies and does not imply contradiction. The regular February 24, 2025 spell (0xd5a30d23) has the highest κ_coord_proxy in the post-2020

sample (238,150 MKR $\times$ 43.6h $\approx$ 10.4M), with HHI=9,931 — a single actor with 99.7% of the MKR. The inversion documented by Villega (2026c) is consistent with this paper's results: HHI does not predict κ_{coord} . What differentiates the February 2025 spell from the subsequent regular spell is not the concentration level, but that the former required active coordination among multiple actors, while the latter was a single actor operating on the prevailing GSM delay.

6.3 Connection with Villega (2026b)

Villega (2026b) (§5.4 and §6) documents that keeper 0xc721 captured 97.68% of the ETH liquidated in the Tariff Shock in 67 minutes, with execution infrastructure operating without pre-positioned capital (flash loans). The asymmetry with κ_{coord} is structural: in the liquidations layer, speed reduces the dominant actor's cost (executes in 67 minutes, no capital immobilized); in the governance layer, speed reduces κ_{coord} only if the spell approves quickly — otherwise, the long window multiplies the cost. Both layers exhibit the same extreme concentration result ($n \approx 1$ in liquidations, $n=2$ in PAs) but through opposite mechanisms: speed in one, inertia in the other.

7. Limitations

Imperfect proxy. $\kappa_{\text{coord_proxy}} = \text{mkr_mobilized} \times \text{gsm_delay_hours}$ captures the opportunity cost of capital immobilized during the mandatory institutional period, but not the organization cost among holders (off-chain communication, timing coordination, spell failure risk). The K^* residual $\approx$ 2.83M DAI includes both dimensions; the proxy captures the first.

Unexplained variance ($R^2=0.672$). The drivers identified in M1–M3 — $\log(1+\text{MKR})$, gsm_delay_hours and $\log(1+n_{\text{lock_events}})$ — explain 67.2% of $\log_{\text{kappa}}$ variance ($R^2=0.672$, M1). The remaining 32.8% reflects unobservable heterogeneity across spells: political importance of the proposal, contextual urgency (out-of-schedule vs. regular spells), and informal coordination among holders leaving no direct on-chain trace. $R^2=0.982$ in M4–M5 reflects sample selection effects toward high participation ($N=149$), not a genuine improvement in explanatory power.

Single sample. The 217 spells are from MakerDAO's DSChief (pre-Sky migration sample). The applicability of the result — that gsm_delay_hours and mkr_mobilized dominate HHI — to other token-governance protocols (Compound, Uniswap, Aave) requires re-estimation under those architectures. In Compound, governor Bravo imposes a minimum quorum of 400,000 COMP, which alters the functional relationship between participation and coordination cost.

Causal identification. The OLS specification with HC3 SEs is correlational. The causal direction $\text{gsm_delay} \rightarrow \kappa_{\text{coord}}$ is the natural hypothesis (the institutional delay forces capital immobilization), but causal reversal is theoretically possible in the mkr_mobilized dimension: a high-mobilization spell may approve faster because the proposing actor already has capital ready. Causal identification would require exogenous variation in gsm_delay_hours uncorrelated with spell characteristics — for example, the legislated delay changes in DSPause documented in Villega (2026c) (§4).

Regime heterogeneity in the temporal sample. The N=217 sample covers 2019–2025 spells, a period that includes structurally distinct GSM delay regimes: ~24h (2019–Mar 2020), emergency reduction to 4h and escalation to 48h (2020), stable 48h (2021–Feb 2023), 16h/48h/16h–30h cycles (2023–2024), and 18h (post-February 2025). The specification includes `gsm_delay_hours` as a continuous variable, capturing within-spell variation, but assumes a stable regression function across six years of heterogeneous regimes. The `post_sky` dummy (n=12, not significant) does not control the pre-2022 high-participation regime with 48h delay — a structurally distinct period from the 2023–2025 regime motivating the connection with Villega (2026a). If the relationship between `gsm_delay_hours` and `κ_coord` varies by regime, the reported coefficients are averages across heterogeneous regimes. Separate estimation by subperiod is statistically risky given low n by regime (~40–60 spells), making this limitation difficult to resolve within the available sample.

Scope: coordination costs, not user welfare. This paper estimates `κ_coord` as a component of Lemma 3's integration cost K, not as a measure of harm to protocol users. The main result — that HHI does not amplify `κ_coord` — implies that voting concentration does not directly raise coordination cost; it does not imply that concentration is irrelevant for depositor welfare. The alternative harm channel operates through risk parameters: a spell approved under compressed window and high concentration can modify collaterals, liquidation ratios or debt ceilings adversely for depositors before they can react with `drop()`. That channel is not mediated by `κ_coord` — it is independent of the spell's coordination cost — and lies outside this paper's scope. Its empirical identification would require a sample of spells with comparable risk parameter changes and depositor behavior data in corresponding delay windows. Villega (2026c) (§8.2) articulates the mechanism for the February 2025 spell as an illustrative case.

8. Series Integration

8.1 Each Paper's Contribution to the Framework

The series shares a single common thread: Lemma 3 of Villega (2026a), which establishes that the maximum number of viable agents n^* in any protocol layer is decreasing in that layer's fixed integration cost K. Each paper instantiates the framework in a different layer, on different data, with different methodologies. The table below synthesizes the four instances.

Table 8.1. Instantiation of Lemma 3 across four papers

Paper	Layer	K interpreted as	Method	Dune Queries	Central Finding	n observed
P1 (2026a)	Deposits (Pot/DSR)	<code>c_gas</code> + <code>c_audit</code> + <code>κ_coord</code> ; $K^* \approx 2.93\text{M DAI}$	S,s model; calibration; OLS	query 7384994	$\lambda \approx 0.86$; $\delta = 0.5703$; $n^* = 2$ (sDAI+Spark, 86% stock)	2

Paper	Layer	K interpreted as	Method	Dune Queries	Central Finding	n observed
			Newey-West			
P2 (2026b)	Liquidations (Clipper)	MEV infrastructure (latency, gas, bots)	HHI; bootstrap; Mann-Whitney MC	ethereum.logs, traces	Tariff Shock: $n^* \approx 1$ (0xc721, 97.68% in 67 min); LUNA: $n^* \approx 399$	≈ 1 under instant shock
P3 (2026c)	Governance (DSChief/GSM)	Outgoing spell coordination cost	31 DuneSQL queries; case analysis	p3_01–p3_31	GSM delay endogenous variable; pre-positioning 10 days before; on-chain access asymmetry	1–4 effective per spell type
P4 (2026d)	Governance (κ_{coord} empirical)	$\kappa_{\text{coord_proxy}} = \text{mkr} \times \text{gsm_delay_hours}$	OLS HC3; N=217 spells, 2019–2025	p4_01, p4_02, p4_03	$\beta(\log \text{MKR}) = 1.13$; $\beta(\text{gsm_delay_hours}) = 0.053$; HHI not significant; $R^2 = 0.672$	— (estimation of K, not n^*)

The trajectory of the four papers is not cumulatively linear: each can be read independently. What makes them a series is that each paper's result feeds back into the others' assumptions. P1 treats GSM delay as a fixed parameter; P3 demonstrates it is an endogenous variable; this modifies the stability of the parameters P1 assumes. P2 treats MEV infrastructure as an external factor; P3 documents its analogue in governance; this generalizes the access asymmetry pattern across the entire protocol.

8.2 The Causal Chain with Empirical Coefficients

The series' central chain, with quantitative values at each link: GSM delay $\uparrow \rightarrow$ mandatory immobilization period $\uparrow \rightarrow \kappa_{\text{coord}} \uparrow \rightarrow K \uparrow \rightarrow n \downarrow$. Each arrow now has an identified empirical anchor.

Arrow 1 — GSM delay $\rightarrow \kappa_{\text{coord}}$: $\beta_2 = 0.053$ (this paper, M1–M3). Per additional hour of GSM delay, $\kappa_{\text{coord_proxy}}$ grows by 5.3%. A spell with delay 48h vs. 12h has $\kappa_{\text{coord_proxy}} \approx e^{(0.053 \times 36)} \approx 6.9$ times larger, all else equal. The prevailing GSM delay at each spell's time is the institutional variable that amplifies coordination cost regardless of who votes.

Arrow 2 — $\kappa_{\text{coord}} \rightarrow K^*$: $\kappa_{\text{coord}} \approx 2.83\text{M DAI} = 96.6\%$ of K^* (Villega (2026a), calibration). The coordination component dominates the other two ($c_{\text{gas}} \approx 800$ DAI, $c_{\text{audit}} \approx 100,000$ DAI) by two orders of magnitude. The empirical estimation in this paper covers the distribution of $\kappa_{\text{coord_proxy}}$ over 217 regular executive spells; Villega

(2026a)'s reference value corresponds to the integration spell establishing $n^*=2$. The connection is a consistency argument, not a direct estimation (§6.1).

Arrow 3 — $K^* \rightarrow n^*$: $n^* \leq D_{\text{total}} \cdot r / (\rho \cdot K)$ (Lemma 3, Villega (2026a)). With $K^* \approx 2.93\text{M DAI}$, $D_{\text{total}} \approx 2,200\text{M DAI}$ and $r=5\%$ DSR, Lemma 3 predicts $n^*=2$, consistent with sDAI and Spark Protocol capturing 86% of stock. In the liquidations layer the same logic produces $n^* \approx 1$ under instant shocks (Villega (2026b), Tariff Shock).

Endogenous amplification mechanism (Villega (2026c)): the 62% decline in spell frequency between 2020 and 2025 implies that each individual spell requires more MKR for longer. This raises κ_{coord} endogenously, without any actor needing to manipulate the delay. The manipulation documented in Villega (2026c) (16h–30h oscillations, reduction to 18h in February 2025) acts as an additional amplifier on a mechanism that already operates structurally.

8.3 Implications for DeFi Protocol Design

The four papers converge on three implications that none can formulate in isolation.

Across the three layers with empirical data — liquidations, governance, and PA deposits — the result is extreme concentration with low n^* . The mechanisms differ: execution speed in liquidations, coordination cost in governance, deployment cost in PAs. But the pattern is consistent enough to suggest that concentration in DeFi protocols is not a correctable design flaw via parametric adjustments, but the structural consequence of markets with heterogeneous fixed costs and asymmetric participation.

P1 and P2 treated the GSM delay as a fixed environmental parameter; P3 demonstrates it is an endogenously contested variable between coalitions with opposing preferences. The implication is not merely empirical: any decentralized governance model that treats GSM parameters as exogenous is formally misspecified, because those parameters are the equilibrium outcome of a control game, not its boundary conditions.

The third implication concerns what 'public' means on-chain. All four papers operate on public Ethereum data — everything documented is technically visible to any participant. The real asymmetry is not data access but the capacity to interpret it in real time: the February 2025 spell's pre-positioning was observable for anyone running P3's 31 queries, and invisible to a depositor or keeper without that analytical infrastructure. Blockchain's technical transparency does not eliminate information asymmetry; it shifts it toward processing speed.

9. Conclusion

This paper empirically estimates κ_{coord} — the coordination component representing 96.6% of the fixed integration cost K^* in Lemma 3 of Villega (2026a) — on 217 executive spells of MakerDAO's DSChief, 2019–2025. In five specifications with HC3 robust standard errors, $\log(1+\text{MKR})$ and gsm_delay_hours are the only statistically significant drivers ($p < 0.001$), with stable coefficients of 1.13 and 0.053 respectively, while voting concentration HHI is not significant in the high-participation subsample ($N=149$).

The result closes two open ends in the series. In Lemma 3 (Villega (2026a)), the cost limiting PA entry is shown to be endogenously increasing in governance spell duration — and that duration increases with the declining participation documented in the same paper, producing a self-reinforcing mechanism that consolidates $n=2$ *without requiring active manipulation*. *In the governance dynamics layer (Villega (2026c)), the correlation between shorter delays and lower κ_{coord} in fast-approval spells is consistent with a differential incentive favoring actors with pre-positioning capacity, though causal identification would require exogenous variation in the delay not available in this sample*. The liquidations result (Villega (2026b)) establishes a structural contrast: extreme concentration in both layers ($n \approx 1$ in liquidations, $n^*=2$ in PAs), through opposite mechanisms — speed reduces the dominant actor's cost in liquidations, while inertia amplifies it in governance. The pattern across all three layers suggests that concentration in decentralized protocols is not a design accident but the structural consequence of any market where fixed costs are heterogeneous and participation declines.

The natural extension is to estimate θ_{out} and θ_{in} of the S,s model directly from variation in κ_{coord} across spell regimes, connecting the on-chain coordination estimation with the PA inertia band parameters — an estimation that would require an observed mass exit regime not available in the 2020–2025 sample.

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Appendix A. Data and Replicability

Table A1. Dune Analytics Queries

Query	ID	Description
p4_01	7503629	Lock events — three DSChief contracts
p4_02	7486415	HHI per spell — reconstruction from raw logs
p4_03	7486567	Master table — subsample N=149 spells with HHI, 2020–2025

Table A2. Reference Contracts

Contract	Address	Period
DSChief v1	0x8e2a84d6ade1e7fffee039a35ef5f19f13057152	2017–2023
DSChief v2	0x9ef05f7f6deb616fd37ac3c959a2ddd25a54e4f5	2019–2026
DSChief (current)	0xa618e54de493ec29432ebd2ca7f14efbf6ac17f7	2020–present

Table A3. Descriptive Statistics — Regression Sample (N=217)

Variable	Mean	Median	SD	Min	Max
κ_{coord} (MKR·h)	215,681	2,197	511,209	0.15	3,035,819
log_kappa	6.14	7.69	4.02	−1.90	14.93
hours_coordination	21.3	12.0	18.7	0.02	84.5
mkr_mobilized	4,821	1,742	8,312	0.01	63,231
hhi	8,149	9,612	2,419	1,207	10,000
n_voters	5.2	4.0	4.8	1	26
whale_present	0.10	0	0.29	0	1
post_sky	0.10	0	0.29	0	1

Table A4. Notation Correspondence Across Series Papers

Concept	Notation P1 (model)	Variable P3 (text/Dune)	Variable P4 (regression)	Name in Dune query
Deposit instrument rate	r_t (continuous process; r_t = DSR between spells)	DSR (discrete value fixed by spell)	r (calibration scalar in Lemma 3: $r = 0.05$)	dsr (maker_ethereum.Pot_call_file)

Concept	Notation P1 (model)	Variable P3 (text/Dune)	Variable P4 (regression)	Name in Dune query
Spread vs. external reference	$s_t = r_t - r_t^*$	DSR/Aave USDC spread	—	spread_dsr_aave (query 7384994)
Governance coordination cost	κ_{coord} (component of K^* ; residual $K^* - c_{\text{gas}} - c_{\text{audit}}$)	kappa_coord (name in text)	κ_{coord} ; $\kappa_{\text{coord_proxy}} = \text{mkr_mobilized} \times \text{gsm_delay_hours}$	—
GSM institutional delay	delay (environment parameter in §4; endogenous variable in P3)	gsm_delay (hours, in text)	gsm_delay_hours (continuous regressor in M1–M5)	gsm_delay_hours (p4_01, query 7503629)
Maximum interval without governance intervention	T_{max} (Prop. 2: $T_{\text{max}} = \min\{\inf\{t: r_t < \theta_{\text{out}}\}, T_{\text{cage}}\}$)	—	—	—

Note: r_t in P1 describes the stochastic process of the spread between discrete governance adjustments; r in P4 is the scalar DSR=5% value used in the Lemma 3 point calibration. They are the same economic variable in two contexts: continuous process (P1) and static calibration value (P4). gsm_delay_hours in P4 corresponds to $\text{ts_exec} - \text{ts_plot}$ of the spell in p4_01, which is the mandatory institutional immobilization period — analogous to the interval P1 calls T_{max} when the delay is not the binding constraint.

Repository: github.com/facundovillega231/defi-research — folder paper4_kcoord

Code: paper4_kcoord/scripts/p4_kcoord_estimation.R

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